Complex Systems Dynamics in Antimalarial Drug Discovery: Polypharmacology, Resistance Networks, and Molecular Neural Network Expressivity Limits

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Malaria drug discovery confronts two intertwined complex systems: the multi-target polypharmacology network of *Plasmodium falciparum* resistance-associated proteins, and the high-dimensional chemical space of African natural product scaffolds exhibiting non-trivial topological complexity. This work bridges statistical physics, stochastic dynamics, and machine learning theory to address three fundamental questions: (1) How does stochastic configurational dynamics (molecular dynamics simulations) diverge from static potential-energy estimates (molecular docking) under mutation-induced perturbations? (2) Can topological descriptors (persistent homology) rescue graph neural network failures on complex molecular structures? (3) What are the expressivity limits of learned representations under distribution shift?

We construct a 65,856-molecule library via scaffold-based generation anchored to 396 African natural products, revealing 92.6% global chemical novelty yet 69.3% conservation of privileged pharmacophores (indoloquinolines, protoberberine alkaloids, quassinoids) [1]. Polypharmacology profiling across four resistance-associated targets identifies 17 multi-target candidates stratified by a Resistance Retention Score (RRS). Molecular dynamics validation (10 ns explicit-solvent trajectories, MM-GBSA free energies) uncovers **87.5% estimand divergence**: static docking predicts attenuated mutant binding, yet short-timescale dynamics show retention in 7/8 systems [2].

Graph neural network benchmarks on 19,836 molecules demonstrate ECFP4 dominance under scaffold split (AUC 0.830 vs GIN 0.805), with topological feature projection modestly rescuing performance on high-Fsp^3 polycyclic structures [3]. 1-Weisfeiler-Leman expressivity bottlenecks emerge on complex natural products, resolvable through distance-aware conformal triage combining nearest-neighbor Tanimoto distance and expected calibration error. The interplay between chemical topology, stochastic transport, and network learning limits defines a frontier where statistical-physics frameworks directly inform resource-constrained drug discovery.

[1] M.V. Sao Temgoua, J.-P. Tchapet Njafa, P. Samafou, F.W. Mbacham, S.G. Nana Engo. Computational discovery of antimalarial candidates from African natural product chemical space. *ChemRxiv* (2026). DOI: 10.26434/chemrxiv.15006437/v1

[2] M.V. Sao Temgoua, J.-P. Tchapet Njafa, P. Samafou, F.W. Mbacham, S.G. Nana Engo. Estimand divergence: static docking versus dynamics for antimalarial triage. *ChemRxiv* (2026). DOI: 10.26434/chemrxiv.15009345/v2

[3] M.V. Sao Temgoua, J.-P. Tchapet Njafa, P. Samafou, F.W. Mbacham, S.G. Nana Engo. Scaffold-controlled evaluation of molecular representations for antimalarial activity prediction. *ChemRxiv* (2026). DOI: 10.26434/chemrxiv.15008517/v1